Notes: Ho and Lee (Econometrica, 2017)
Insurer Competition in Health Care Markets

Contents

1 Big picture

- **Question.** How does insurer competition affect premiums, negotiated hospital prices, hospital utilization, and surplus in a health-care market where insurers bargain with hospitals and also bargain or set premiums with the employer?
- **Empirical setting.** The paper studies the California Public Employees' Retirement System (CalPERS), where households choose among Blue Shield (BS), Blue Cross (BC), and Kaiser plans across 14 hospital service areas (HSAs).
- **Core challenge.** Removing an insurer has two equilibrium effects that move in opposite directions: fewer insurers can raise premiums, but the remaining insurers may gain bargaining leverage over hospitals, lowering hospital prices.
- **Model structure.** Households choose insurers, patients choose hospitals conditional on insurance networks, insurers negotiate premiums with the employer, and insurers bargain bilaterally with hospitals over admission prices.
- **Key institutional feature.** CalPERS does not just passively accept insurer premiums. The model allows premium setting to be constrained by employer-insurer bargaining, which is essential for matching observed margins and counterfactual premium responses.
- **Main counterfactual.** Removing Kaiser from the market raises Blue Shield premiums by about 16.6% and raises Blue Cross premiums by about 14.4%; removing Blue Cross lowers Blue Shield premiums by about 3.4%.
- **Hospital-price result.** Removing Kaiser raises Blue Cross hospital prices by about 19.9%, while removing Blue Cross lowers Blue Shield hospital prices by about 8.9%.
- **Mechanism.** The premium effect tends to increase negotiated hospital prices because higher insurer revenues increase gains from trade with hospitals. The enrollment effect tends to lower prices because a larger remaining insurer is more valuable to hospitals. The net sign depends on which insurer is removed.
- **Economic takeaway.** More insurer competition does not mechanically lower all health-care prices. Competition among insurers can reduce premiums, but insurer size also shapes bargaining leverage over providers. A policy that changes insurer market structure must account for both sides of the market.

2 Incremental contribution of this paper

- **Beyond reduced-form insurer concentration regressions.** Earlier empirical work often linked concentration to premiums or provider prices separately. This paper estimates a structural model that jointly determines insurance demand, hospital demand, premiums, and hospital-insurer bargaining outcomes.
- **Equilibrium counterfactuals with two bargaining margins.** The paper's distinctive contribution is to quantify how removing an insurer changes both downstream premium competition and upstream hospital bargaining. These two channels can have opposite signs.
- **Employer bargaining over premiums.** A central innovation is the treatment of CalPERS as an active negotiator over premiums. The paper shows that assuming Nash-Bertrand premium setting can predict counterfactual premium movements that differ sharply from the bargaining model.
- **Micro-founded hospital price decomposition.** The model decomposes changes in negotiated hospital prices into premium/enrollment effects, price reinforcement, hospital cost, and recapture effects. This makes the counterfactual mechanism transparent.
- **Insurer identity matters.** The paper moves beyond generic "more or fewer insurers" analysis. Removing a high-share integrated insurer such as Kaiser has very different implications from removing a smaller or less attractive insurer.

- **Policy relevance.** The framework is directly relevant to health-care antitrust and exchange design. It shows that consumer-facing competition and provider-facing bargaining power must be evaluated jointly rather than separately.

3 Data and measurement

3.1 Institutional setting and products

- The study focuses on CalPERS enrollees in California, who choose among BS, BC, and Kaiser plans.
- BS and BC are modeled as managed care organizations with hospital networks and negotiated hospital prices.
- Kaiser is vertically integrated and has a more restricted hospital system. It is not available in all HSAs.
- Premiums vary by plan and household type: single, two-party, and family.

Interpretation: The setting is attractive because households face a limited menu of plans, insurers have distinct hospital networks, and the market has enough cross-HSA variation to estimate demand and bargaining primitives.

3.2 Hospital and household data

- The paper uses CalPERS enrollment and claims data, hospital discharge information, and hospital cost measures.
- Hospitals are grouped into hospital systems, and hospital concentration is summarized using system membership and admission-based HHI.
- Admission probabilities and DRG weights vary by age-sex cells and are used to predict hospital utilization and payments.
- Hospital prices are DRG-adjusted prices per admission, constructed from total payments divided by DRG-weighted admissions for each hospital-insurer pair.

Interpretation: The DRG adjustment is important because observed payments reflect case mix. The model needs a hospital price measure that captures negotiated prices rather than severity differences across admissions.

3.3 Demand objects

- The model estimates individual hospital choice conditional on plan network and health event.
- The model also estimates household insurance plan choice as a function of premiums, plan characteristics, and expected hospital utility.
- Household plan demand generates own-premium elasticities for each household type and plan.

Interpretation: Insurance demand and hospital demand are linked: a plan's network affects expected hospital access and therefore household willingness to pay for the plan.

3.4 Bargaining objects

- The model estimates insurer non-hospital marginal costs, insurer-hospital Nash bargaining parameters, and the premium bargaining parameter.
- Hospital-insurer bargaining uses the loss in insurer premium revenue and the loss in hospital admissions upon disagreement.
- Employer-insurer premium bargaining is captured by a parameter that constrains premiums relative to pure Nash-Bertrand competition.

Interpretation: The bargaining estimates are the bridge from demand primitives to counterfactual prices. Without this bridge, a change in insurer competition cannot be translated into hospital price changes.

4 Models and empirical specifications

4.1 Hospital demand

The utility of individual k, with diagnosis l, enrolled in plan j, and living in market m, from visiting hospital i is modeled as

$$u^H_{kijlm} = \delta_i + z_i v_{kl} \beta^z + d_{ik} \beta^d_m + \varepsilon^H_{kijlm}. \tag{1}$$

Hospital choice is constrained by the individual's hospital market and by the insurer's network.

Interpretation: The hospital demand system converts networks into expected household utility and predicts how admissions move when an insurer or hospital is removed.

4.2 Insurance plan demand

Households choose insurance plans based on premiums, plan characteristics, market-insurer fixed effects, and expected hospital utility. A stylized representation is

$$u^M_{hjm} = \alpha_p p_{hjm} + \xi_{jm} + \Omega^H_{hjm} + \varepsilon^M_{hjm}, \tag{2}$$

where p_{hjm} is the premium, ξ_{jm} captures plan-market attractiveness, and Ω^H_{hjm} summarizes expected hospital utility.

Interpretation: This specification allows the model to predict enrollment shifts after an insurer leaves the choice set and premiums adjust.

4.3 Hospital-insurer bargaining

For each hospital-insurer pair (i, j), the negotiated price balances the gains from trade for the insurer and the hospital. The model implies a first-order condition that depends on

- the insurer's loss in premium revenue if hospital i is removed from its network,
- the hospital's loss in admissions if insurer j is removed,
- the negotiated price itself,
- hospital costs,
- insurer-hospital bargaining weights.

Interpretation: Hospital prices are not exogenous inputs. They respond to changes in insurer enrollment, premiums, and hospital outside options.

4.4 Premium setting and premium bargaining

The model compares a pure Nash-Bertrand premium-setting benchmark with a specification in which CalPERS bargaining constrains premiums. A stylized premium first-order condition can be written as

$$\frac{\partial \Pi_j}{\partial p_j} = 0 \quad \text{under Nash-Bertrand competition,} \tag{3}$$

whereas the bargaining model allows premiums to reflect employer-insurer bargaining over surplus.

Interpretation: Matching observed insurer margins requires accounting for premium bargaining. This is why the paper's main specification differs from a standard differentiated-products Bertrand model.

4.5 Counterfactual decomposition

Hospital price changes are decomposed into components:

$$\Delta p^H = \text{premium/enrollment effect} + \text{price reinforcement} + \text{cost effect} + \text{recapture effect}. \tag{4}$$

Interpretation: This decomposition is the key to understanding why the same reduction in insurer competition can raise prices in one case and lower prices in another.

5 Identification and empirical design

5.1 What identifies demand

- Hospital demand is identified from patient choices among hospitals within networks, conditional on diagnosis, demographics, distance, hospital fixed effects, and networks.
- Plan demand is identified from household enrollment choices across insurers, household types, markets, and premiums.
- Market-insurer fixed effects help separate price sensitivity from persistent insurer-market attractiveness.

Interpretation: The model uses both the structure of choice sets and cross-market variation in insurer networks and premiums to recover demand primitives.

5.2 What identifies bargaining parameters

- Negotiated hospital prices, hospital costs, hospital demand, and insurer demand jointly discipline hospital-insurer bargaining parameters.
- Insurer margins discipline the premium-bargaining parameter and non-hospital marginal costs.
- Bootstrap procedures account for estimation uncertainty in hospital price construction and bargaining parameters.

Interpretation: The bargaining side is identified from how observed prices compare to what hospitals and insurers would lose under disagreement.

5.3 Why the counterfactual is credible

- The counterfactuals re-solve equilibrium rather than applying partial-equilibrium price changes.
- Removing an insurer changes consumer choice sets, enrollment, premiums, hospital bargaining outside options, and hospital prices simultaneously.
- The paper reports confidence intervals from recomputing counterfactuals under bootstrap samples.

Interpretation: The results are not just correlations between concentration and prices. They are simulated equilibrium outcomes from an estimated demand-and-bargaining model.

5.4 Limitations

- The counterfactuals hold hospital and insurer networks fixed except for the removed insurer.
- Long-run entry, merger, network redesign, benefit design, and provider capacity responses are outside the main model.
- Kaiser is vertically integrated, so its price and network structure are treated differently from BS and BC.
- The estimated model is tailored to CalPERS and may not mechanically generalize to all health-insurance markets.

Interpretation: The paper's strength is the equilibrium mechanism; the main caution is that it is a short-run structural counterfactual in a particular institutional environment.

6 Tables and figures

6.1 Figure 1: Predicted premium and hospital-price changes, and Table I: Summary statistics

Read:

- Figure 1 compares the removal of Blue Cross and Kaiser under two premium-setting assumptions.
- Under unconstrained Nash-Bertrand premium competition, removing Blue Cross raises Blue Shield premiums by 11.0%, while removing Kaiser raises them by 19.3%.
- Under the bargaining specification, removing Blue Cross lowers Blue Shield premiums by 3.4%, while removing Kaiser raises them by 16.6%.
- Hospital prices also differ sharply by which insurer is removed. Removing Kaiser raises hospital prices for Blue Shield by 0.6%, while removing Blue Cross lowers Blue Shield hospital prices by 8.9%.
- Table I shows that BS, BC, and Kaiser differ in premiums, hospital networks, negotiated prices, and enrollment.
- BS and BC have much broader networks than Kaiser. BS has 189 hospitals in network; BC has 223; Kaiser has 27.

Economic message: Figure 1 is the paper's preview of the central mechanism: insurer competition affects both premiums and provider bargaining. Table I shows why insurer identity matters; insurers differ in market shares, networks, prices, and costs.

	(a) Premium Changes: Insurer Removed		(b) Hospital Price Changes: Insurer Removed	
Premium Setting	BC (Small)	K (Large)	BC (Small)	K (Large)
Unconstrained (Nash–Bertrand)	+11.0% [10.8%, 11.3%]	+19.3% [19.1%, 19.6%]	−1.1% [−1.5%, −0.8%]	+3.0% [2.1%, 3.9%]
Constrained (Bargaining)	−3.4% [−4.0%, −3.3%]	+16.6% [15.8%, 16.8%]	−8.9% [−13.3%, −7.7%]	+0.6% [−3.1%, 1.8%]

FIGURE 1.—Predicted (a) premium and (b) hospital price per admission changes for Blue Shield upon the removal of either Blue Cross (BC) or Kaiser (K), when insurers set premiums according to Nash–Bertrand competition or bargain with the employer. 95% confidence intervals are reported below estimates. See Section 4 for details.

(a) Figure 1: Predicted premium and hospital-price changes

TABLE I

SUMMARY STATISTICS[a]

		BS	BC	Kaiser
Premiums (per year)	Single	3782.64	4192.92	3665.04
	2-Party	7565.28	8385.84	7330.08
	Family	9834.84	10,901.64	9529.08
	Revenues (per individual)	2860.34	3179.39	2788.05
Insurer Characteristics	# Hospitals in Network	189	223	27
	# Hospital Systems in Network	119	149	–
	Hospital Prices (per admission)	7191.11	6023.86	–
	Hospital Payments (per individual)	623.20	554.00	–
	Hospital Costs (per admission)	1709.56	1639.92	–
Household Enrollment	Single	19,313	8254	20,319
	2-Party	16,376	7199	15,903
	Family	35,058	11,170	29,127
	Avg. # Individuals/Family	3.97	3.99	3.94

[a]Summary statistics by insurer. The number of hospitals and hospital systems in network for BS and BC are determined by the number of in-network hospitals or systems with at least 10 admissions observed in the data. Hospital prices and costs per admission are average unit-DRG amounts, weighted across hospitals by admissions. Hospital payments per individual represent average realized hospital payments made per enrollee (not weighted by DRG).

(b) Table I: Summary statistics

Adjacent visuals: Figure 1 and Table I

6.2 Table II: Individual enrollment and hospital system concentration, and Table III: Admission probabilities and DRG weights

Read:

- Table II shows large cross-market variation in enrollment and hospital concentration.
- BC is small in many markets but has a large share in some rural markets; Kaiser is unavailable in HSAs 1 and 8.
- Los Angeles has many hospitals and low concentration, while smaller markets often have fewer hospital systems and higher HHI.

- Table III shows admission probabilities by age-sex categories. Older age groups have much higher admission probabilities and higher DRG weights.
- Women aged 20–34 have high admission probabilities but relatively low DRG weights, consistent with labor and delivery admissions.

Economic message: The two tables document the two demand layers: insurance enrollment varies across markets, while hospital admission risk varies across patient types. Both matter for the bargaining value of an insurer's network.

TABLE II

INDIVIDUAL ENROLLMENT AND HOSPITAL SYSTEM CONCENTRATION[a]

	Individual Plan Enrollment						Hospital Concentration			
	Enrollment			Market Share			# Systems		HHI (Adm)	
HSA Market	BS	BC	Kaiser	BS	BC	Kaiser	BS	BC	BS	BC
1. North	5366	15,143	–	0.26	0.74	–	5	17	3686	1489
2. Sacramento	55,732	6212	59,772	0.46	0.05	0.49	6	8	4112	2628
3. Sonoma / Napa	6826	955	13,762	0.32	0.04	0.64	5	5	3489	3460
4. San Francisco Bay West	6021	926	4839	0.51	0.08	0.41	4	4	4362	3054
5. East Bay Area	7856	1200	10,763	0.40	0.06	0.54	9	10	2560	2096
6. North San Joaquin	9663	3979	4210	0.54	0.22	0.24	7	8	2482	1888
7. San Jose / South Bay	2515	762	4725	0.31	0.10	0.59	5	6	3265	2628
8. Central Coast	8028	13,365	–	0.38	0.62	–	4	9	3431	2254
9. Central Valley	27,663	7613	10,211	0.61	0.17	0.22	12	13	1863	1539
10. Santa Barbara	3973	1416	658	0.66	0.23	0.11	7	7	2459	2863
11. Los Angeles	18,205	6731	23,919	0.37	0.14	0.49	22	28	741	716
12. Inland Empire	17,499	2801	20,690	0.43	0.07	0.50	15	15	1015	1034
13. Orange	7836	2906	5430	0.48	0.18	0.34	8	9	2425	2250
14. San Diego	14,585	2298	8593	0.57	0.09	0.34	10	8	1708	2549
Total[b]	191,768	66,307	167,572	0.45	0.16	0.39	119	147	1004	551

[a]Individual enrollment and market shares (Kaiser was not an option for enrollees in HSAs 1 and 8) and hospital system membership and admission Herfindahl–Hirschman Index (HHI)—computed using the number of admissions for all hospital-insurer pairs in our sample—by insurer.

[b]Total (statewide) HHI accounts for hospital system membership across HSAs.

(a) Table II: Enrollment and hospital concentration

TABLE III

ADMISSION PROBABILITIES AND DRG WEIGHTS[a]

	Admission Probabilities			DRG Weights		
	OSHPD	CalPERS		CalPERS		
Age–Sex Category	All	BS	BC	BS	BC	All
0–19 Male	2.05%	1.78%	2.08%	1.78	1.49	1.70
20–34 Male	2.07%	1.66%	2.07%	1.99	1.77	1.92
35–44 Male	3.11%	2.79%	3.21%	1.95	1.89	1.93
45–54 Male	5.58%	5.29%	5.32%	2.07	2.05	2.07
55–64 Male	10.49%	10.13%	9.70%	2.25	2.25	2.25
0–19 Female	2.28%	1.95%	2.04%	1.31	1.39	1.32
20–34 Female	11.19%	11.75%	10.22%	0.84	0.87	0.85
35–44 Female	7.91%	7.31%	7.73%	1.32	1.33	1.32
45–54 Female	6.87%	6.16%	6.82%	1.90	1.83	1.87
55–64 Female	9.74%	9.01%	9.26%	2.03	2.02	2.03

[a]Average admission probabilities and DRG weights per admission by age–sex category. OSHPD refers to

(b) Table III: Admission probabilities and DRG weights

Adjacent visuals: Table II and Table III

6.3 Table IV: Insurance plan household price elasticities, and Figure 2: Premiums by premium Nash bargaining parameter

Read:

- Table IV reports own-premium elasticities for each plan and household type.
- Demand is more elastic for family plans than for single plans, with family elasticities around −2.5 to −3.0.
- The elasticities are substantial enough that premium changes can cause meaningful enrollment substitution.
- Figure 2 plots predicted single household premiums as the premium bargaining parameter varies.
- Premiums rise strongly as the premium bargaining parameter increases toward the Nash-Bertrand benchmark, especially for Blue Cross.
- The steepness at high parameter values explains why the premium-setting assumption materially changes counterfactual predictions.

Economic message: These visuals show that premium competition is economically important and that the employer's role in constraining premiums is central to the model's predictions.

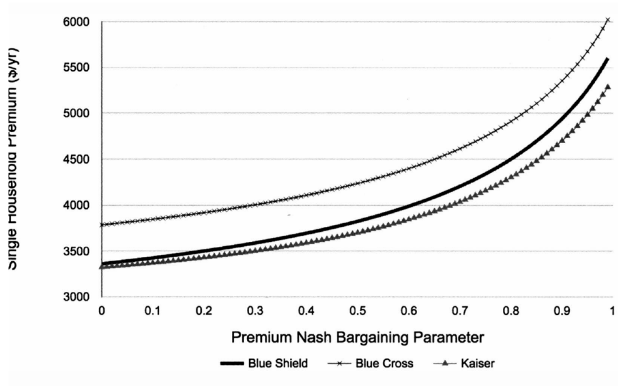
TABLE IV

ESTIMATES: INSURANCE PLAN HOUSEHOLD PRICE ELASTICITIES[a]

	Single	2-Party	Family
BS	−1.23	−2.15	−2.53
BC	−1.62	−2.50	−2.95
Kaiser	−1.23	−2.12	−2.53

[a]Estimated own-price elasticities for each insurer using insurer demand estimates from Table A.IV.

(a) Table IV: Household price elasticities



(b) Figure 2: Premiums and premium bargaining

Adjacent visuals: Table IV and Figure 2

6.4 Table V: Insurer marginal costs and bargaining parameters, and Table VI: Negotiated hospital price decomposition

Read:

- Table V reports two specifications. Specification (i) assumes Nash-Bertrand premiums; specification (ii) uses margin moments and allows employer-insurer premium bargaining.
- In specification (ii), non-hospital marginal costs are higher: about 1691 for BS, 1949 for BC, and 2535 for Kaiser.
- The estimated premium bargaining parameter is 0.47 in specification (ii), indicating that premiums are constrained below the pure Nash-Bertrand benchmark.
- Table VI decomposes negotiated hospital prices for BS and BC.
- The largest determinant of hospital prices is the price reinforcement effect: 66.3% for BS and 52.6% for BC.
- Premium and enrollment effects also matter substantially, while hospital cost and recapture effects are smaller.

Economic message: Table V identifies the pricing regime, and Table VI explains why hospital prices respond to premium and enrollment changes. Together, they show that insurer competition operates through both downstream and upstream channels.

TABLE V

ESTIMATES: INSURER MARGINAL COSTS AND NASH BARGAINING PARAMETERS[a]

		(i)	(ii)
Insurer Non-Inpatient	η_{BS}	925.78	1691.50
Marginal Costs		11.12	10.41
(per individual)	η_{BC}	1417.73	1948.61
		6.93	8.14
	η_{K}	1496.44	2535.14
		–	0.62
Nash Bargaining	τ_{BS}	0.33	0.31
Parameters		0.01	0.05
	τ_{BC}	0.40	0.38
		0.02	0.03
	τ^{ϕ}	1.00	0.47
		–	0.00
Use Margin Moments		N	Y
Number of Bilateral Pairs		268	268

[a]2-step GMM estimates of marginal costs for each insurer (which do not include hospital payments for BS and BC), Nash bargaining parameters, and elasticity scaling parameter. When "margin moments" are not used, we set $\tau^{\phi} = 1.00$, and Kaiser marginal costs are directly obtained from (12) by setting $\omega^{1}_{\text{Kaiser}} = 0$. Standard errors are computed using 80 bootstrap samples of admissions

(a) Table V: Marginal costs and bargaining parameters

TABLE VI

ESTIMATES: NEGOTIATED HOSPITAL PRICE DECOMPOSITION[a]

	Price	(i) Premium & Enrollment	(ii) Price Reinforcement	(iii) Hospital Costs	(iv) Recapture Effect
BS	7191.11	24.2%	66.3%	8.9%	0.6%
		[23.6%, 25.5%]	[64.9%, 69.3%]	[5.1%, 10.6%]	[0.4%, 0.8%]
BC	6023.86	32.3%	52.6%	12.1%	3.0%
		[31.8%, 33.7%]	[51.8%, 55.1%]	[9.2%, 13.1%]	[2.3%, 3.3%]

[a]Weighted average (by hospital admissions) decomposition of negotiated hospital prices into the components provided in (A.3) for each insurer and hospital system (omitting residuals, and scaling by τ_j or $1 - \tau_j$ where appropriate). 95% confidence intervals, reported below estimates, are constructed using 80 bootstrap samples of admissions within each hospital-insurer pair to re-estimate hospital-insurer DRG weighted admission prices, re-estimate insurer marginal costs and Nash bargaining parameters, and re-compute

(b) Table VI: Hospital price decomposition

Adjacent visuals: Table V and Table VI

6.5 Table VII: Removing an insurer, summary results, and Table VIII: Hospital price changes across markets

Read:

- Table VII reports the main counterfactual equilibrium effects of removing Kaiser or removing Blue Cross.
- Removing Kaiser increases BS premiums by 16.6% and BC premiums by 14.4%, and raises BC hospital prices by 21.2%.
- Removing Blue Cross lowers BS premiums by 3.4%, lowers BS hospital prices by 8.9%, and reduces hospital surplus.
- Table VIII shows that hospital price effects vary substantially across markets.
- With premiums fixed, removing Kaiser generally reduces hospital prices because remaining insurers gain enrollment and bargaining leverage.
- Allowing premiums to adjust can reverse the sign: higher premiums increase gains from trade and can raise negotiated hospital prices.
- The decomposition columns reveal that the premium effect and enrollment effect often work in opposite directions.

Economic message: These are the paper's main policy results. The market power effect of insurer concentration and the bargaining-power effect of insurer size are both present; the net effect depends on insurer identity and equilibrium premium adjustment.

REMOVING AN INSURER: SUMMARY RESULTS[a]

		Baseline	(i) Remove Kaiser		(ii) Remove BC	
		Amount	Amount	% Change	Amount	% Change
Premiums (per year)	BS	3.78	4.41	16.6%	3.65	−3.4%
		[3.76, 3.79]	[4.36, 4.43]	[15.8%, 16.8%]	[3.62, 3.66]	[− 4.0%, −3.3%]
	BC	4.19	4.80	14.4%	–	–
		[4.18, 4.20]	[4.75, 4.81]	[13.7%, 14.6%]		
	Kaiser	3.67	–	–	3.62	−1.4%
		[3.66, 3.67]			[3.60, 3.62]	[− 1.6%, −1.3%]
Household Enrollment	BS	73.91	124.16	68.0%	87.73	18.7%
		[73.65, 74.34]	[124.13, 124.25]	[67.1%, 68.6%]	[87.44, 88.51]	[18.4%, 19.3%]
	BC	27.49	38.56	40.2%	–	–
		[27.49, 27.50]	[38.47, 38.59]	[39.9%, 40.4%]		
	Kaiser	61.31	–	–	64.99	6.0%
		[60.88, 61.58]			[64.21, 65.27]	[5.2%, 6.3%]
Hospital Payments (per individual)	BS	0.66	0.66	0.5%	0.60	−8.5%
		[0.65, 0.68]	[0.64, 0.68]	[− 3.1%, 1.7%]	[0.57, 0.62]	[− 12.7%, −7.5%]
	BC	0.56	0.68	21.2%	–	–
		[0.55, 0.58]	[0.67, 0.72]	[20.0%, 24.8%]		
Hospital Prices (per admission)	BS	7.19	7.23	0.6%	6.55	−8.9%
		[7.06, 7.35]	[6.92, 7.43]	[− 3.1%, 1.8%]	[6.19, 6.74]	[− 13.3%, −7.7%]
	BC	6.02	7.29	21.0%	–	–
		[6.04, 6.40]	[7.14, 7.64]	[19.8%, 24.6%]		
Surplus (per individual)	Insurer	0.44	0.99	125.9%	0.38	−13.3%
		[0.44, 0.44]	[0.99, 0.99]	[124.6%, 126.6%]	[0.38, 0.39]	[− 13.8%, −11.7%]
	Hospitals (Non-K)	0.30	0.51	69.7%	0.27	−9.0%
		[0.29, 0.31]	[0.49, 0.52]	[63.0%, 72.3%]	[0.26, 0.28]	[− 13.8%, −7.6%]
	Δ Cons.	–	−0.19	–	−0.01	–
			[− 0.19, −0.18]		[− 0.01, −0.01]	

[a]Results from simulating removal of Blue Cross or Kaiser from all markets using estimates from specification (iv) in Table V. All figures are in thousands. Baseline numbers (including premiums, hospital prices, and enrollment) are recomputed from model estimates. Average insurer payments to hospitals and average DRG-adjusted hospital prices are weighted by the number of admissions each hospital receives from each insurer under each scenario. Surplus figures represent total insurer, hospital, and changes to consumer surplus per insured individual. 95% confidence intervals, reported below estimates, are constructed by using 80 bootstrap samples of admissions within each hospital-insurer pair to re-estimate hospital-insurer DRG weighted admission prices, re-estimate insurer marginal costs and Nash bargaining parameters, and re-compute counterfactual simulations.

(a) Table VII: Removing an insurer - summary results

REMOVING AN INSURER: COUNTERFACTUAL BLUE SHIELD AND BLUE CROSS HOSPITAL PRICE CHANGES ACROSS MARKETS[a]

	Avg. Hospital Price ($/Admission)					Decomposition of Change ($/Admission)				
		Fix Premiums		Adjust Premiums						
	Baseline	CF	% Change	CF	% Change	(ia) Prem Effect	(ib) Enroll Effect	(ii) Price Reinforce	(iii) Cost Effect	(iv) Re-Capture
(ia) REMOVE KAISER: BS PRICES										
All Mkts	7191.13	6451.01	−10.29%	7175.65	−0.22%	624.97	−1149.39	473.70	0.65	34.59
2. Sacramento	8204.98	7318.75	−10.80%	7751.96	−5.52%	605.39	−1572.02	491.33	1.83	20.45
4. SF Bay W.	8825.62	7994.95	−9.41%	8589.65	−2.67%	616.37	−1439.98	533.81	−0.86	54.69
5. E. Bay	7368.50	5967.77	−19.01%	6537.55	−11.28%	717.37	−1820.40	229.04	0.15	42.89
9. C. Valley	6591.73	6369.72	−3.37%	7329.03	11.19%	556.42	−550.32	681.83	0.00	49.36
10. S. Barbara	7934.89	7779.92	−1.95%	8709.83	9.77%	402.15	−187.53	533.88	2.55	23.90
11. LA	5878.37	4829.25	−17.85%	5661.03	−3.70%	662.05	−1163.77	258.83	0.43	25.12
14. SD	6673.04	6038.49	−9.51%	6634.70	−0.57%	472.14	−908.62	380.01	−0.04	18.16
(ib) REMOVE KAISER: BC PRICES										
All Mkts	6023.83	5988.53	−0.59%	7219.85	19.85%	671.85	−130.41	580.01	0.24	74.33
2. Sacramento	6651.31	6703.09	0.78%	8186.10	23.08%	839.58	−137.89	728.48	2.05	102.58
4. SF Bay W.	7602.06	7734.73	1.75%	9189.30	20.88%	836.40	−157.26	747.50	−0.70	161.29
5. E. Bay	7158.45	7150.76	−0.11%	8570.60	19.73%	835.46	−220.00	684.32	0.18	112.19
9. C. Valley	5210.75	5215.51	0.09%	6763.68	29.80%	875.55	−134.94	700.05	0.00	112.27
10. S. Barbara	5130.74	5094.60	−0.70%	6395.60	24.65%	699.55	−84.34	599.56	2.52	47.55
11. LA	6084.19	5803.18	−4.62%	6960.25	14.40%	687.32	−386.22	540.62	0.21	34.12
14. SD	5381.70	5482.36	1.87%	6841.04	27.12%	807.95	−143.63	719.75	−0.02	75.29
(ii) REMOVE BLUE CROSS: BS PRICES										
All Mkts	7191.13	6898.64	−4.07%	6620.28	−7.94%	−129.81	−247.77	−167.38	0.01	−25.89
2. Sacramento	8204.98	8098.96	−1.29%	7799.41	−4.94%	−125.74	−131.81	−134.28	−0.02	−13.72
4. SF Bay W.	8825.62	8643.19	−2.07%	8370.37	−5.16%	−128.03	−195.86	−95.34	0.10	−36.12
5. E. Bay	7368.50	7252.44	−1.58%	6913.99	−6.17%	−149.00	−113.83	−170.56	0.00	−21.11
9. C. Valley	6591.73	5945.62	−9.80%	5781.16	−12.30%	−115.57	−485.97	−152.72	−0.02	−56.29
10. S. Barbara	7934.89	7248.92	−8.65%	7170.32	−9.64%	−83.53	−610.90	−17.78	−0.28	−52.08
11. LA	5878.37	5623.27	−4.34%	5304.90	−9.76%	−137.51	−216.72	−200.27	−0.02	−18.94
14. SD	6673.04	6373.32	−4.49%	6161.37	−7.67%	−98.07	−239.34	−160.35	0.00	−13.91

[a]Average (DRG-adjusted) hospital prices for Blue Shield from simulating the removal of Blue Cross or Kaiser across all HSAs, or within a selected sample of HSAs, using estimates from specification (iv) in Table V. Baseline numbers are recomputed from model estimates. Average hospital prices are weighted by the number of admissions each hospital receives from each insurer under each scenario. Decomposition effects correspond to terms in equation (A.4), and are weighted by the number of admissions under the baseline scenario; their sum equals the predicted overall change in hospital prices.

(b) Table VIII: Hospital price changes across markets

Adjacent visuals: Table VII and Table VIII

6.6 Table IX: Removing an insurer under Nash-Bertrand premium setting

Read:

- Table IX repeats the Blue Cross removal scenario under Nash-Bertrand premium setting.
- Unlike the main bargaining specification, Nash-Bertrand predicts that Blue Shield premiums rise by 11.0% and Kaiser premiums rise by 8.7% when Blue Cross is removed.
- Hospital prices for Blue Shield fall slightly, by 1.1%, and insurer surplus rises sharply by 24.1%.
- The comparison with Table VII shows how sensitive policy conclusions are to the premium-setting model.

Economic message: Table IX is a diagnostic table. It shows that ignoring CalPERS premium bargaining can lead to qualitatively different conclusions about the impact of insurer exit.

TABLE IX

REMOVING AN INSURER: SUMMARY RESULTS (NASH–BERTRAND PREMIUM SETTING)[a]

		Baseline	(iii) Remove BC (Nash–Bertrand)	
		Amount	Amount	% Change
Premiums (per year)	BS	3.78 [3.76, 3.79]	4.20 [4.17, 4.22]	11.0% [10.8%, 11.3%]
	BC	4.19 [4.18, 4.21]	–	–
	Kaiser	3.67 [3.66, 3.67]	3.98 [3.97, 4.00]	8.7% [8.4%, 8.9%]
Household Enrollment	BS	73.91 [73.53, 74.56]	82.99 [82.71, 83.39]	12.3% [11.8%, 12.5%]
	BC	27.49 [27.06, 27.77]	–	–
	Kaiser	61.31 [61.10, 61.44]	71.13 [70.78, 71.38]	16.0% [15.8%, 16.2%]
Hospital Payments (per individual)	BS	0.66 [0.65, 0.68]	0.66 [0.65, 0.67]	−0.4% [−0.7%, −0.1%]
	BC	0.56 [0.55, 0.58]	–	–
Hospital Prices (per admission)	BS	7.19 [7.06, 7.36]	7.11 [6.96, 7.29]	−1.1% [−1.5%, −0.8%]
	BC	6.02 [6.03, 6.40]	–	–
Surplus (per individual)	Insurer	1.27 [1.27, 1.27]	1.57 [1.57, 1.58]	24.1% [23.4%, 24.7%]
	Hospitals (Non-K)	0.30 [0.29, 0.31]	0.29 [0.28, 0.30]	−2.8% [−3.9%, −1.9%]
	Δ Cons.	–	−0.09 [−0.09, −0.08]	–

[a]Results from simulating removal of Blue Cross or Kaiser, using estimates from specification (i) in Table V (without insurer margin moments) and assuming Nash–Bertrand premium setting. All figures are in thousands. Baseline numbers are recomputed from model estimates. Average insurer payments to hospitals and average (DRG-adjusted) hospital prices are weighted by the number of admissions each hospital receives from each insurer under each scenario. Surplus figures represent total insurer, hospital, and changes to consumer surplus per insured individual. 95% confidence intervals, reported below estimates, are constructed by using 80 bootstrap samples of admissions within each hospital-insurer pair to re-estimate hospital-insurer DRG weighted admission prices, re-estimate insurer marginal costs and Nash bargaining parameters, and re-compute counterfactual simulations.

Table IX: Removing an insurer under Nash-Bertrand premium setting

7 Short summary

- Ho and Lee estimate a structural model of health-insurance competition that jointly determines insurer demand, hospital demand, premiums, and hospital-insurer bargaining.
- The empirical setting is CalPERS in California, where households choose among Blue Shield, Blue Cross, and Kaiser across 14 HSAs.
- The model captures two-sided effects of insurer competition: consumer-facing premium competition and provider-facing bargaining power.
- CalPERS premium bargaining is essential for matching observed insurer margins and counterfactual premium changes.
- Removing Kaiser increases premiums for the remaining insurers and raises Blue Cross hospital prices substantially.
- Removing Blue Cross lowers Blue Shield premiums and lowers Blue Shield hospital prices in the main bargaining specification.
- Hospital price changes are decomposed into premium/enrollment effects, price reinforcement, cost, and recapture effects.
- The premium effect and enrollment effect can offset each other, which is why counterfactual signs differ across insurer removals.
- The paper shows that policies affecting insurer competition cannot be evaluated solely by looking at premiums or provider prices in isolation.

8 Expanded conclusion

8.1 Core conclusion

The paper's central conclusion is that insurer competition has ambiguous equilibrium effects once both premium competition and hospital bargaining are modeled. Fewer insurers can increase premiums by reducing consumer-facing competition, but larger remaining insurers can also extract lower prices from hospitals. Which force dominates depends on the insurer removed, the insurer's attractiveness, hospital networks, market concentration, and the premium-setting institution.

8.2 Incremental contribution revisited

The paper adds to the health-care industrial organization literature by embedding hospital-insurer bargaining inside an estimated insurance demand system and then solving full equilibrium counterfactuals. Its incremental contribution is not just another estimate of insurer market power. It is the decomposition of insurer competition into downstream premium effects and upstream bargaining effects. This decomposition makes the model useful for antitrust and public policy because it clarifies why insurer consolidation may lower negotiated provider prices while raising or lowering premiums depending on the institutional setting.

8.3 Economic intuition

The intuition is that an insurer is valuable to hospitals only because it delivers patient volume. Removing a rival insurer changes the patient volume and premium revenue of the remaining insurers. When a remaining insurer becomes more attractive or captures more enrollment, hospitals may need access to that insurer more, increasing the insurer's bargaining power. But if premiums rise, the insurer's revenue from having a hospital in-network also rises, which increases the gains from agreement and can raise hospital prices. The counterfactual outcome is therefore the net of two opposing forces.

8.4 Policy implications

The findings caution against one-dimensional measures of competition. A merger or exit can reduce the number of insurers, but the welfare effect depends on whether premium increases dominate hospital-price reductions. The paper also highlights the importance of employers and plan sponsors. If large employers can constrain premiums, the impact of insurer competition differs from what a standard Nash-Bertrand model would predict. For regulators, the lesson is to evaluate insurers as intermediaries between consumers and providers, not merely as sellers of insurance contracts.

8.5 Limitations and interpretation

The results are structural counterfactuals in a specific institutional setting. They are not reduced-form causal estimates of actual insurer exits. They also focus on short-run equilibrium with fixed hospital networks and benefit design. Long-run insurer entry, hospital investment, network redesign, and consumer plan switching frictions could alter the magnitudes. Nevertheless, the model provides a clear framework for organizing these forces and shows that ignoring either premium bargaining or hospital bargaining can lead to misleading conclusions.

8.6 Takeaway

The broad takeaway is that insurer competition is a two-sided market problem. More insurers can discipline premiums, but fewer or larger insurers can strengthen bargaining power against hospitals. The paper's main achievement is to quantify these competing forces in one equilibrium model and show that their interaction is large enough to reverse policy conclusions.